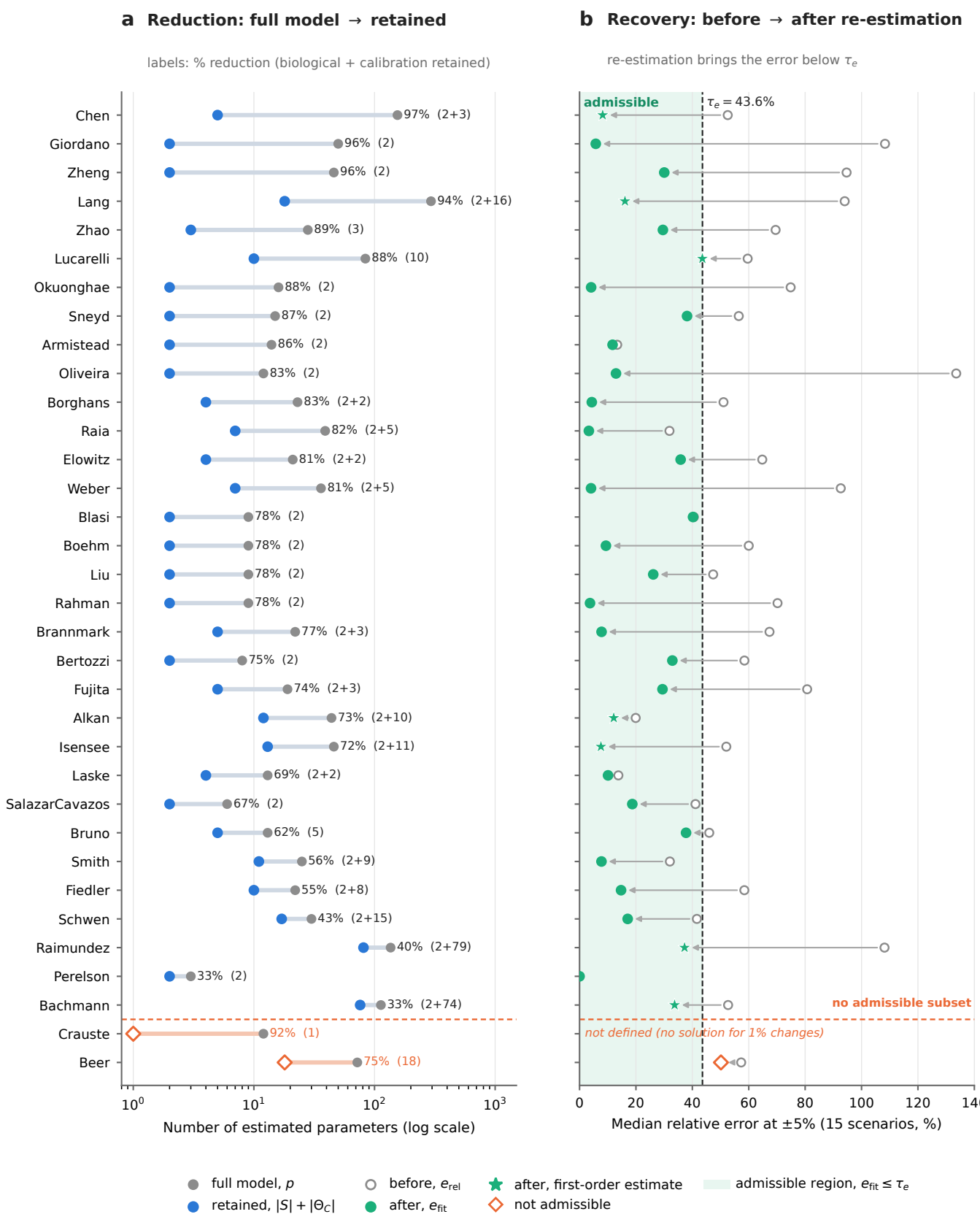

Parameter reduction and recovery of the full-model output in 34 benchmark problems
a Reduction: full model → retained
labels: % reduction (biological + calibration retained)
Chen 97% (2+3)
Giordano 96% (2)
Zheng 96% (2)
Lang 94% (2+16)
Zhao 89% (3)
Lucarelli 88% (10)
Okuonghae 88% (2)
Sneyd 87% (2)
Armistead 86% (2)
Oliveira 83% (2)
Borghans 83% (2+2)
Raia 82% (2+5)
Elowitz 81% (2+2)
Weber 81% (2+5)
Blasi 78% (2)
Boehm 78% (2)
Liu 78% (2)
Rahman 78% (2)
Brannmark 77% (2+3)
Bertozzi 75% (2)
Fujita 74% (2+3)
Alkan 73% (2+10)
Isensee 72% (2+11)
Laske 69% (2+2)
SalazarCavazos 67% (2)
Bruno 62% (5)
Smith 56% (2+9)
Fiedler 55% (2+8)
Schwen 43% (2+15)
Raimundez 40% (2+79)
Perelson 33% (2)
Bachmann 33% (2+74)
Crauste 92% (1)
Beer 75% (18)
10^0 10^1 10^2 10^3
Number of estimated parameters (log scale)
b Recovery: before → after re-estimation
re-estimation brings the error below τ_e
admissible
$\tau_e = 43.6\%$
no admissible subset
not defined (no solution for 1% changes)
0 20 40 60 80 100 120 140
Median relative error at ±5% (15 scenarios, %)
full model, p
retained, $|S| + |\Theta_C|$
before, e_{rel}
after, e_{fit}
after, first-order estimate
not admissible
admissible region, $e_{fit} \le \tau_e$